

Balireddy Mahalakshmi

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Career Objective

AI/ML Graduate with a strong foundation in building intelligent systems that solve real-world problems. I specialize in developing deep learning models—like YOLO and LSTMs—and connecting them to live data sources. I have hands-on experience in Python automation, predictive analytics, and deploying solutions using AWS cloud. I'm passionate about making AI practical, scalable, and easy to use in everyday life.

Technical Skills

- **Programming Languages:** Python, SQL (MySQL), C, Data Structures & Algorithms (DSA).
- **Machine Learning:** Supervised/Unsupervised Learning, Deep Learning (CNN, LSTM), NLP.
- **Libraries & Frameworks:** TensorFlow, Keras, OpenCV, Pandas, NumPy, Matplotlib.
- **Tools & Cloud:** AWS (Certified), FastAPI, Git, GitHub, Power BI, VS Code, MS Excel.

Professional Experience

Machine Learning Intern | FabHost Tech Solutions | Jan 2024 – Mar 2024

- Developed and optimized **supervised models** (Random Forest, XGBoost) using Python and Scikit-learn, achieving a 10% increase in accuracy through advanced feature engineering.
- Automated **data-cleaning pipelines** by developing custom Python scripts, reducing manual data preparation time by 20% and ensuring high data integrity.
- Streamlined ML workflows by implementing modular code and version control, improving the scalability and reproducibility of team-wide experiments.

Projects

Netra: AI-Based Real-Time Assistive Vision System

- Developed a **multi-modal AI pipeline** integrating **YOLOv8** for object detection, **EasyOCR** for text recognition, and **MobileNetV2** for high-accuracy currency classification.
- Architected a **Client-Server infrastructure** using **Android (CameraX)** and **FastAPI**, enabling real-time frame capture and high-speed model inference via a modular backend.
- Optimized **system performance** using **OpenCV** for image preprocessing and **Asynchronous API calls (Retrofit)** to deliver seamless, low-latency audio feedback via Text-to-Speech (TTS).

IoT-Based Weather Forecasting System

- Architected a **real-time data pipeline** using **ESP32** sensors and **TensorFlow**, developing an **LSTM neural network** for high-accuracy time-series weather forecasting.
- Engineered an **interactive dashboard** by integrating **Google Maps API**, enabling real-time location-based data visualization and sensor mapping.
- Developed **automated alerting workflows** using **IFTTT protocols** and asynchronous data streams to provide instant mobile notifications based on environmental thresholds.

Education

Bachelor of Technology (B.Tech) in AI & Machine Learning

Saveetha Engineering College | 2021 – 2025 | **CGPA: 8.0/10**

Intermediate, MPC

Sri Chaitanya Junior College | 2019 – 2021 | **Percentage: 80%**

Certifications

- AWS Certified Cloud Practitioner | Amazon Web Services
- Machine Learning with Python | IBM / Udemy
- Python Programming Certification | FabHost Tech Solutions